

# Oleksii Krasnoshtanov

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Thermo Fisher Scientific  
Materials & Structural Analysis, Engineering  
Eindhoven, Netherlands

## Re: Gen AI Assistant Internship, Engineering — Eindhoven

Dear Hiring Team,

I am writing about the Gen AI Assistant internship with the Engineering team in Eindhoven — profiling Docgraph for retrieval time, evaluating existing guardrail implementations, reducing hallucinations, and moving a proof of concept into the MSD R&D environment.

In 2025 I built a self-hosted retrieval assistant for MARBET, a corporate events agency, in a two-person sprint; they run it in production. It answers from the client's own confidential documents, which is why it runs on-premise: Llama 3.3 70B (q5\_K\_M) through Ollama, orchestrated with LangChain over mxbai-embed-large embeddings in FAISS. Most of the effort went into answer quality rather than the model. I chunked semantically on paragraph and then sentence boundaries instead of fixed cuts, used two-stage fetch-then-filter retrieval, returned the source passages with every answer, and made the assistant say plainly when the documents did not cover a question rather than fill the gap. A guided assistant for Smart EPU has the same failure mode with higher stakes: a confident wrong answer about which squares or foil holes to collect from costs a user a microscope session, while an honest “not covered” costs them a click.

The on-prem constraint is part of why the work appeals to me. Your software lives on customer infrastructure — Embedded CryoSPARC Live on the microscope hardware, Athena inside an existing on-premises environment — and a pharma customer's structures are not going to a third-party API. I have shipped into that shape: on an MLOps platform for NPEC I ran one codebase to both on-premise Portainer and Azure Container Apps, with blue-green cutover and health-gated rollback. Getting a proof of concept to survive the move into a controlled R&D environment is a specific skill, and it is the one I expect would be most immediately useful.

Where I would want to start is measurement: a labelled question set over the real documentation, retrieval recall and latency measured separately from generation, so that every later change to chunking, reranking or guardrails can be argued with a number instead of an impression. It is also the prerequisite for the hallucination work.

I am available from February 2027 and would be glad to talk through any of this.

Kind regards,  
Oleksii Krasnoshtanov